

Unit - 1

ASS-1

Short Answer Test

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1.)

a.) Classification of OS and Subclassification

Type : Multi-user, Multitasking operating system

Subclassification :

- > Supports multiple users at the same time
- > Executes multiple processes simultaneously.
- > Shares CPU, memory, and I/O devices efficiently.
- > Provides security and user isolation.

b.) Structure of an operating system

Users

|
Application program

|
System calls (API)

|
kernel

|
Process management

|
memory management

File management
|
Device management
|
Hardware.

1.) System call sequence

fork()
exec()
wait()
exit()

Eg:

- 1.) Parent creates child using fork()
- 2.) Child loads program using exec()
- 3.) Parent waits using wait()
- 4.) Child finishes using exit()

2.)

a.) System calls for File and Device management

File management

- open()
- read()
- write()
- close()
- lseek()

Device management

- `ioctl()`
- `read()`
- `write()`
- `open()`
- `close()`

b) Layered Architecture

User program

File System

Device Drivers

Hardware

c) C system call sequence

```
int fd;
```

```
char buffer[100];
```

```
fd = open ("Sample.txt", O_RDONLY);
```

```
read (fd, buffer, size of (buffer));
```

```
close (fd);
```

3.)

a.) Suitable OS

Real-time operating System (RTOS)

- Fast response Time
- Meets deadlines
- Predictable execution
- Used in embedded and industrial systems.

b.) Interaction Diagram

User process

|

kernel

|

Scheduler

|

memory manager

|

Device driver

|

Hardware

c.) Synchronization System calls

• `fork()`

• `wait()`

• `Sem_wait()`

• `Sem_post()`

• `pthread_mutex_lock()`

a.) Types of OS and comparison

Distributed OS

Multiple computers

Shares resources

High reliability

Fault tolerance

Centralized OS

Single computer

Local resources only

Less reliable

Limited processing

b.) Distributed OS Architecture

Node 1

|

Network

|

Node 2

|

Network

|

Node 3

|

Shared Resources

c.) IPC System Call sequence

pipe()

fork()

write()

read()

close()
wait()

5)

a) Major Functions of OS

- 1.) Process scheduling
- 2.) Memory management
- 3.) Device management.
- 4.) File management.
- 5.) Security and protection.

b.) Block diagram

Users

↓
Applications

↓
Operating System

↓
Process management
Memory management

File management

Device management

Security

↓
Hardware